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Development of a solar irradiance cloud index model for Northwest Argentina using GOES-16 imagery. Comparison against reanalysis and satellite databases.

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Abstract: Satellite databases provide critical climatic and environmental information, covering extensive geographical regions and long temporal scales. These databases primarily provide meteorological data and solar estimates that can be used to conduct preliminary assessments of the feasibility of installing solar power plants. Still, the accuracy of these assessments depends on evaluating the errors associated with the model estimates from each database. However, the models are fixed, limiting their ability to be tailored to specific regional or local conditions. To overcome this limitation, it is feasible to develop a custom satellite model based on the cloudiness index (CIM) method, which requires constructing a background reflectance map from satellite imagery. Initial CIM models have been developed using GOES-16 satellite data for Argentina's Salta and Jujuy provinces. For the calibration of these local models, ground-based measurements of global solar irradiance have been used from three different climate sites. To evaluate the performance of these local models, their solar irradiance estimates were compared with those provided by reanalysis databases. The locally calibrated CIM models outperform the satellite and reanalysis models, suggesting that developing a CIM model specific to Argentina's northern region could significantly improve current estimates.

1. INTRODUCTION

Using satellite images to estimate solar radiation values over large regions, such as countries or continents, is a handy tool for determining the spatial and temporal characteristics of the resource [1]. This does not imply that measuring solar irradiance at the surface is not essential, but quite the contrary: stations with quality instruments are necessary to monitor the models' performance using satellite images to calculate their estimates [2]. By comparing the measured values against those estimated by satellite models, we know if those models are adequate for a region because no model describes precisely what happens at the surface [3]. Thus, knowing the error of a satellite database (SDB) is very important when using satellite information to perform, for example, pre-feasibility calculations for solar power plant installation.

Interestingly, the SDB offers global solar irradiance (GHI) as a variable and presents others associated with meteorological weather, such as temperature, relative humidity, wind direction, and speed. Associating this with the Normalized Difference Vegetation Index (NDVI) [4], the SDB is also a source of environmental information.

For the north of Argentina, there are four open-access SDBs: CAMS-Rad (model HELIOSAT-4), LSA-SAF (model MDSSFTD), NSRDB (model PSM-2), and GOES-DSR (model DSR). All are hourly, and some are even sub-hourly. These databases have provided valuable information on the region's characteristics and have been used to compare values measured at sites in the area and perform the Site Adaptation procedure [5]. This technique allows for generating a function that relates the measured solar irradiance values and those estimated by the SDB with the same time stamp, i.e., simultaneous. This period where the adaptation function is sought is called the simultaneity period. Once this function is found, it can be applied to the SDB estimates values outside the simultaneity period, where no measured data are available. In this way, a much larger time series can be obtained than the one available only using measured values, and an increase in the statistical representativeness of the variable over time can be achieved. However, the models mentioned above are closed, so it is not possible to improve them or adapt them to regional conditions in a straightforward way.

However, developing a regional version of a Cloud Index Model (CIM) is possible. Thus, using values measured on the ground and analyzing satellite images over the region of interest (GOES-16 in this case), a model that allows us to know the values of solar irradiance almost in real time can be developed.

In this work, we analyze the results by comparing the GHI values measured at three sites in northern Argentina to those estimated using a local CIM model adjusted to each site. We have used GOES-16 images to develop this CIM model, preprocessing them to extract the required information. The results indicate that the local CIM model is better than its open-access counterparts.

2. GROUND MEASUREMENTS

GHI data measured at three sites in northwestern Argentina (see Table 1) were used to calibrate the CIM. These sites were chosen because they have different climates, according to the Koppen-Geiger classification [6], which gives the CIM model its versatility. Reanalyses are not as accurate as satellite-based estimates [7], so we will only compare the measured values and those estimated by ERA-5 for other periods (see Table 2)

Table 1

Characteristics of the three sites analyzed. Note the altitude difference between locations.

Site (Province)	Climate Type	Latitude S (°)	Longitude W (°)	Altitude asl (meters)	Sensor	Period
Yuto (Jujuy)	Cwa	23.5845	64.5066	401	K&Z CMP11	2017
Salta (Salta)	Cwb	24.7288	65.4095	1233	Eppley PSP	2023
El Rosal (Salta)	ET	24.3928	65.7681	3355	K&Z CMP3	2018

Table 2

Metrics for the complete dataset of measured GHI vs. those estimated by CAMS-Rad and ERA5 for each site.

Site (Province)	# hours	rMBE CAMS	rRMSE CAMS	rMBE ERA5	rRMSE ERA5
Yuto (Jujuy)	5373	0.55	24.38	-1.01	36.03
Salta (Salta)	6502	6.53	35.24	10.95	40.59
El Rosal (Salta)	8940	-24.43	40.43	-14.25	28.04

Known methods have covered the most common performance indicators in solar resource assessment. [24]; these include the Mean Bias Error (MBE), Mean Absolute Error (MAE), and the Root Mean Square Error (RMSE). The three metrics are defined as follows

$$MBE = \frac{\sum_{i=1}^{n} x_i - y_i}{n} \quad (1)$$

$$MAE = \frac{\sum_{i=1}^{n} |x_i - y_i|}{n} \quad (2)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (x_i - y_i)^2}{n}} \quad (3)$$

where x and y are the measured and estimated values, respectively, and n is the sample size. The MBE measures the systematic bias that a model can introduce in a long-term evaluation. In contrast, the MAE and RMSE measure the error dispersion using absolute and quadratic norms, respectively. Because of its greater sensitivity to outliers, the RMSE is often used in this area. Both dispersion metrics are reported here for completeness. The three indicators are presented in relative terms as a percentage of the mean of the measured values, referred to here as rMBE, rMAE, and rRMSE.

3. ANALYZED SATELLITE AND REANALYSIS MODELS

Four SDBs with free-access data for the northwestern region of Argentina have been mentioned. Each will be described below.

3.1 CAMS HELIOSAT-4

Heliosat-4 [8] is an entirely physical model that employs a rapid yet precise approach to the libRadtran radiative transfer model [9]. This model utilizes various satellite data and the reanalysis database from the Copernicus Atmosphere Monitoring Service (CAMS) reanalysis database. Cloud properties are derived from Meteosat Second Generation (MSG) satellite images at a 15-minute time rate using an adapted APOLLO scheme (AVHRR3 Processing Scheme Over Clouds, Land, and Ocean). The operational version of Heliosat-4 takes the form of lookup tables, enabling rapid computation. This model has already been evaluated in the region [10, 11]; however, these studies need to report a comparison with other modeled solar estimation data sets.

3.2 NREL NSRDB

The National Solar Radiation Database (NSRDB) uses the Physical Solar Model (PSM) [12]. This two-stage model integrates cloud properties from GOES satellites (4×4 km, 30-minute intervals), aerosol optical depth from MODIS and MERRA-2 (0.5° resolution, interpolated to 4 × 4 km), and surface albedo from the MODIS MCD43GF product (30 arcseconds, 8-day intervals). The PSM uses two radiative transfer models: REST2 for clear sky conditions and FARMS (Fast All-Sky Radiation Model for Solar applications) [13] for all sky conditions. FARMS uses precalculated cloud transmittances and reflectances parameterized as functions of solar zenith angle, cloud phase, optical thickness, and particle size. This approach makes FARMS about 1000 times faster than traditional two-stream methods while maintaining comparable accuracy. Data processing includes re-gridding inputs to 4 x 4 km resolution and gap filling for missing cloud properties.

3.3 GOES DSR

The GOES Downward Shortwave Radiation (DSR) product, developed by NESDIS/NOAA, uses the Enterprise Processing System (EPS) Shortwave Radiation Budget (SRB) hybrid algorithm to estimate incident solar radiation reaching the Earth's surface and reflected solar radiation exiting the top of the atmosphere from visible and near-infrared GOES satellite imagery. This process involves several steps, including sensor data calibration, atmospheric correction for clear sky cases, and determination of the amount of clouds and aerosols present in the atmosphere [14]. The SRB algorithm is run at the pixel level, assigning each pixel to one of four categories: clear sky without snow/ice, clear sky with snow/ice, water cloud, and ice cloud. Two estimation paths are used: direct if the required Level 2 (L2) input data is available and indirect otherwise. The direct path adapts the CERES [15] model using precomputed lookup tables based on the Fu and Liou RTM, considering L2 satellite products such as surface albedo, aerosol optical depth, and aerosol single scattering albedo for clear-sky scenes, and cloud optical depth, radius, and top height for cloudy scenes. The indirect path uses the climatological values of aerosol optical depth and single scattering albedo for the given date if the pixel is clear sky one. If the pixel is cloudy, the atmospheric transmission is estimated using the top of the atmosphere albedo derived from the imager reflectances. This model in the region of interest has a spatial resolution of 0.5 ° in latitude and longitude and a temporal update frequency of 60 minutes.

3.4 LSA-SAF MDSSFTD

The MSG Downwelling Surface Short-wave Radiation Fluxes - Total and Diffuse (MDSSFTD) product is an instantaneous (every 15 minutes) estimate of the surface's global and diffuse downward shortwave radiation flux. The method for retrieving this product consists of two separate modules: one for clear sky and one for cloudy sky conditions [16, 17]. This is a physical parameterization method coupled to a pre-calculated lookup table (LUT) of aerosol properties, mainly of the direct and diffuse transmittances of the different aerosol components and their corresponding aerosol albedos. The LUT is generated using radiative transfer models to vary aerosol loading, water vapor, and solar zenith angle. The cloudy GHI retrieval method uses the total shortwave cloud albedo at the top of the atmosphere (TOA) obtained from the observed TOA reflectances of the MSG. It uses the same aerosol estimates as in the clear sky case, and cloud-related terms are determined using a simplified radiative transfer model. Finally, the effective clear-sky transmittance estimated using gases and aerosols and the cloud transmittance estimated using the simplified radiative transfer model are used together to determine the GHI. This model has an approximate spatial resolution of 3 km with a 15-minute update frequency.

3.5 ERA-5

ERA5 is the fifth-generation atmospheric reanalysis model developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), covering the period from January 1940 to the present. This model provides hourly estimates of numerous climate variables, including atmospheric, terrestrial, and oceanic aspects. The data are presented on a horizontal grid of 30 km, and the atmosphere is binned using 137 levels ranging from the Earth's surface to an altitude of 80 km. The model is based on the Integrated Forecasting System (IFS) Cy41r2, operational until 2016. Thus, ERA5 benefits from a decade of advances in model physics, core dynamics, and data assimilation compared to the previous ERA-Interim data set. In addition to a significantly improved horizontal resolution of 0.25° latitude (31 km), ERA-5 provides hourly output and uncertainty estimates derived from the set (every 3 hours at half the horizontal resolution). Using advanced modeling and data assimilation techniques, this model combines satellite and in-situ observations in its estimates; therefore, ERA-5 has produced more accurate simulations than previous generations [18].

4. THE CIM MODEL.

The satellite-based model tested in this work is locally implemented. It is a Cloud Index Model (CIM) applied using visible channel images from the GOES-16 satellite [19]. The model formulation is:

$$GHI = GHI_{cs}\ x\ F(\eta) \qquad (4)$$

$$\eta = \frac{\rho - \rho_{min}}{\rho_{max} - \rho_{min}} \qquad (5)$$

where GHI_{cs} are clear sky estimates from the Argpv2 model [20], F is a cloud attenuation function, and η is the cloud index [21]. The value of ρ_{max} was set to 80%, as in Refs. [22], and ρ_{min} was calculated as the average of the twenty minimum values of ρ in a 300-hour moving window centered on the point of interest as proposed in Ref. [23]. The cloud attenuation function F(η) is defined as (a ∗ (1 - η) + b). The Argpv2 clear-sky model has a local adjustment for the study region of this work.

In this work, we have compared the values measured at the stations against those estimated by only two satellite databases: CAMS and LSA-SAF. NSRDB only has data from 2019 to 2022, and GOES-DSR estimates are only from 2020. The CIM model developed is called GCIM for GOES-CIM.

5 RESULTS

Below, the values generated by the CIM model for each site are compared to those estimated by the CAMS and LSA-SAF databases. Figure 1 shows the overall results, expressed as rRMSE.

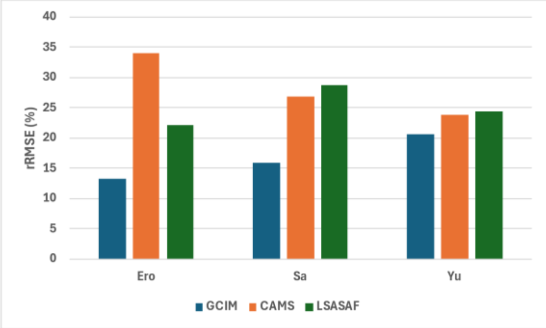

Fig. 1 Comparison of the rRMSE metric between the estimated GHI values at each site, between the GCIM and the SDBs.

Fig. 1 shows that the GCIM model improves the estimates at each site compared to the values estimated by the SDBs used. The information at each site varies when comparing by solar zenith angle ranges. We have discriminated by solar zenith angle size range to increase the comparison detail between the values estimated by the GCIM at each site. This is shown in Figures 2, 3, and 4.

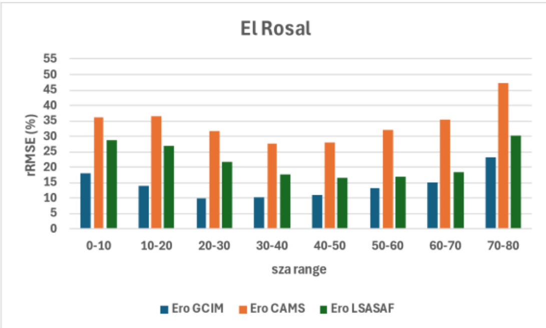

Fig.2 Comparison of the rRMSE metric between the estimated GHI values at El Rosal, between the GCIM and the SDBs, using sza ranges.

The information at each site varies when compared by solar zenith angle ranges. In the case of El Rosal (Fig.2), CAMS has a significant error for LSA-SAF. The local GCIM model is better in each zenith angle range.

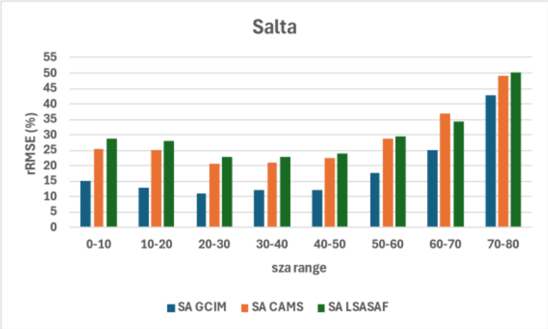

Fig.3 Comparison of the rRMSE metric between the estimated GHI values at Salta, between the GCIM and the SDBs, using sza ranges

In Salta (Fig.3), the CAMS and LSA-SAF estimates are similar, and the GCIM estimates are better than those already mentioned.

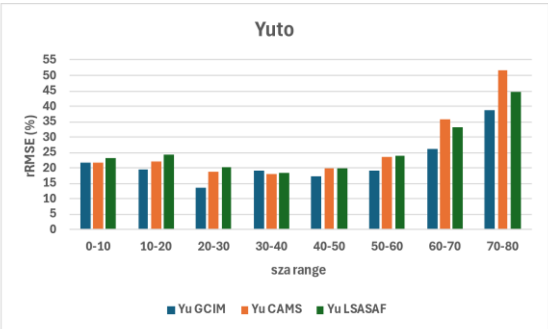

Fig.4 Comparison of the rRMSE metric between the estimated GHI values at Yuto, between the GCIM and the SDBs, using sza ranges

The values for each range are similar for Yuto (Fig.4), indicating that the CAMS and LSA-SAF models achieve quality estimates for this site, although with angular problems.

6. CONCLUSIONS

This paper compares GHI estimates for two open-access databases, CAMS-Rad and LSA-SAF, and one generated locally using GOES-16 imagery and calibrating against ground-based values, against measured GHI values.

Initial results show that local GCIM models are better than those available in open-access databases.

Thus, a CIM satellite model for Argentina's northern region could be developed using GOES-16 images and data from radiometric and meteorological ground stations.

This would allow the development of real-time analysis and forecasting tools for solar radiation and environmental information. These tools would help determine the availability of real-time photovoltaic energy generation and other variables of scientific and technical interest. This would influence optimizing areas such as mining, distributed generation, or disaster control.

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